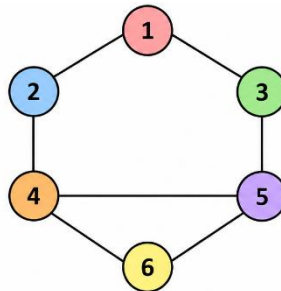


Q1) Write a program to solve a state-space search problem using DFS. **[10 marks]**

Start vertex - 1



Q2) Write Python program to demonstrate the following objective functions used in clustering algorithms. (Any 2 of the following) **[20 marks]**

1. K-Means Inertia (Within-Cluster Sum of Squares – WCSS)
2. Gaussian Mixture Model (GMM) Log-Likelihood
3. Silhouette Coefficient
4. Davies-Bouldin Index

Using matplotlib to plot the Elbow curve, Silhouette scores, and Davies-Bouldin scores across different values of K (on x-axis).

OR

Q2) Write a program to build a Knowledge Base using logical rules and facts. **[20 marks]**

Q3) Viva **[05 marks]**